

Cryptocurrency Price Tracker

PROJECT DESCRIPTION:

The Cryptocurrency Price Tracker is a Selenium-powered Python tool that dynamically scrapes real time cryptocurrency prices from CoinMarketCap . It collects essential details such as coin name, current price, 24-hour change, and market cap.

This project automates browser interaction using Chrome WebDriver to load JavaScript-based pages and extract accurate market data. It is ideal for analysts, investors, and developers seeking real-time data for research or dashboard integration.

FEATURES:

- 1. Live Price Scraping**
Scrapes real-time prices and market data for leading cryptocurrencies.
- 2. Dynamic Page Handling**
Uses Selenium WebDriver to load and extract data from JavaScript-rendered pages.
- 3. Top 10 Coins Data**
Extracts details such as coin name, price, 24h change, and market cap for the top 10 cryptocurrencies.
- 4. CSV Export**
Saves the extracted data into a structured CSV file for further analysis or integration.
- 5. Headless Browser Option**
Supports headless mode for running the script in the background without opening a browser window.
- 6. Historical Logging**
Appends timestamped data to the CSV file to enable trend tracking over time.
- 7. Filtering by Price or Change (Optional)**
Allows filtering coins based on custom conditions like price threshold or highest 24-hour gainers.

TECHNOLOGIES USED:

- 1. Python**
Primary programming language for scraping and automation.
- 2. Libraries**
 - Selenium: For dynamic scraping and browser control.
 - pandas: For data manipulation and export.
 - webdriver_manager: To automatically manage ChromeDriver setup.

- time: For controlling scraping delays and timestamps.

3. Browser Automation

Google Chrome controlled through ChromeDriver.

4. Data Storage

CSV format for storing coin data.

OUTCOMES:

1. Real-Time Market Monitoring
Enables tracking of live cryptocurrency prices and fluctuations.
2. Trend Analysis
Historical data logging supports future trend forecasting and comparative analysis.
3. Custom Alert and Filtering Logic
Allows integration of custom filters for price monitoring or investment decision-making.
4. Dashboard Ready
The exported data can be connected to visualization tools for building dashboards.
5. Portfolio Tracking
Helps investors monitor key metrics of coins in their watchlist.